

RETAIL SALES & INVENTORY ANALYTICS

Retail Sales & Inventory Analytics Dashboard

Power BI Dashboard Development | Data Analytics

An end-to-end analytics solution that turns two years of multi-store retail data into a single, decision-ready view of sales, inventory, and demand performance.

TOOLS USED

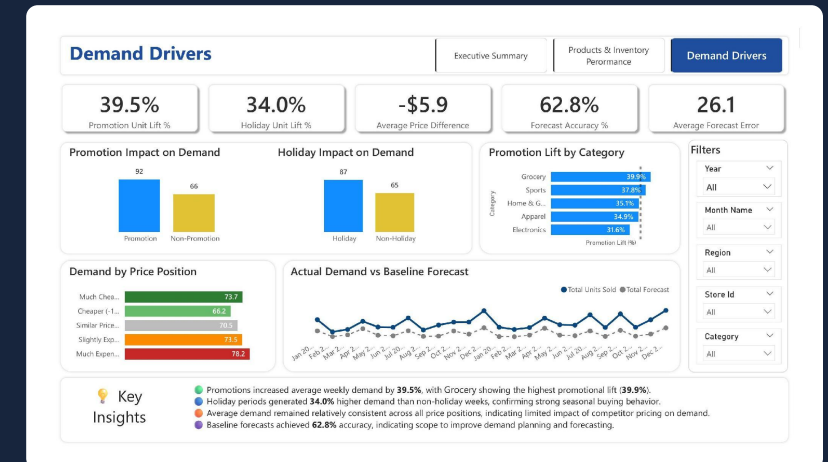
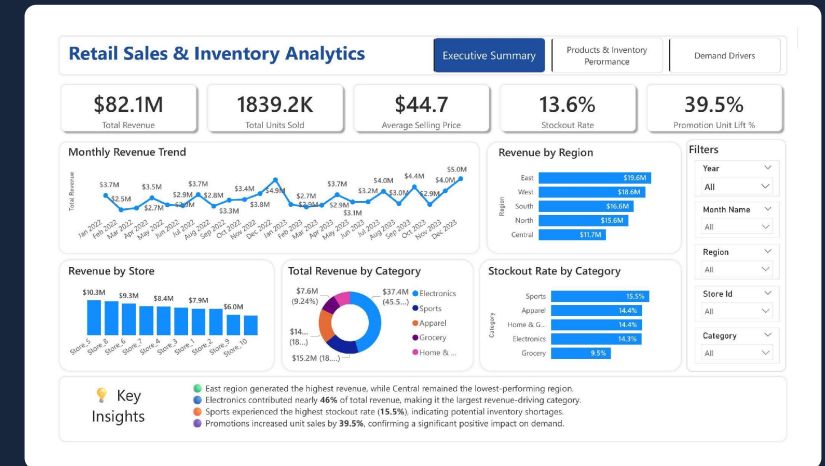
Power BI

DAX

Python

Excel

Power Query



Live interactive Power BI dashboard — 3 report pages, full cross-filtering

THE BUSINESS PROBLEM

Growing Revenue While Avoiding Empty Shelves

Retail businesses must balance sales growth with inventory availability while responding to changing demand, promotions, holidays, and competitor pricing. Without a centralized view, decision-makers struggle to identify revenue opportunities, prevent stockouts, and understand the factors driving demand, resulting in fragmented reporting and slower decisions.



Revenue Visibility

Hard to see what's driving — or dragging down — revenue across regions, stores and categories.



Inventory Shortages

Stockouts go unnoticed until sales are already lost, especially during peak demand periods.



Promotion Effectiveness

Limited visibility into promotion effectiveness, making it difficult to identify high-impact campaigns.



Forecast Gaps

Planning relies on demand forecasts of unknown accuracy, reducing confidence in inventory decisions.

The goal: The goal was to replace fragmented, manual reporting with a single interactive source of truth for sales and inventory performance.

DATASET OVERVIEW

A Two-Year, Multi-Store Retail Dataset

2022–2023, weekly grain · sales, inventory, pricing, promotions, holidays, competitor pricing & demand forecasts



2 Years

of weekly data



10 Stores

across 5 regions



25 Products

across 5 categories



26,250

weekly records

PRODUCT CATEGORIES

- Electronics
- Grocery
- Apparel
- Home & Garden
- Sports

MAIN DATA COLUMNS

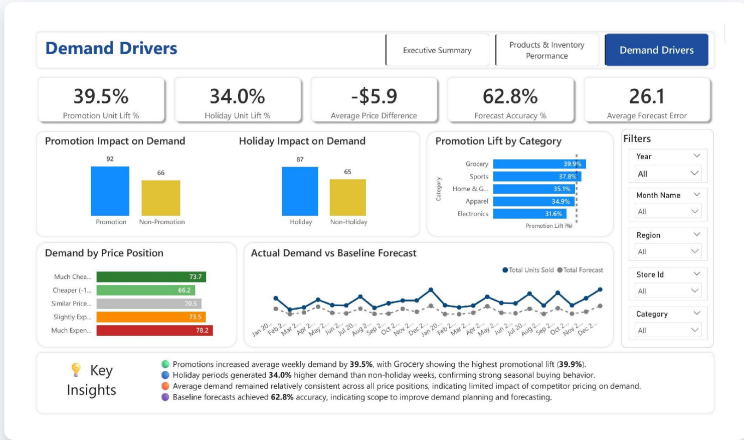
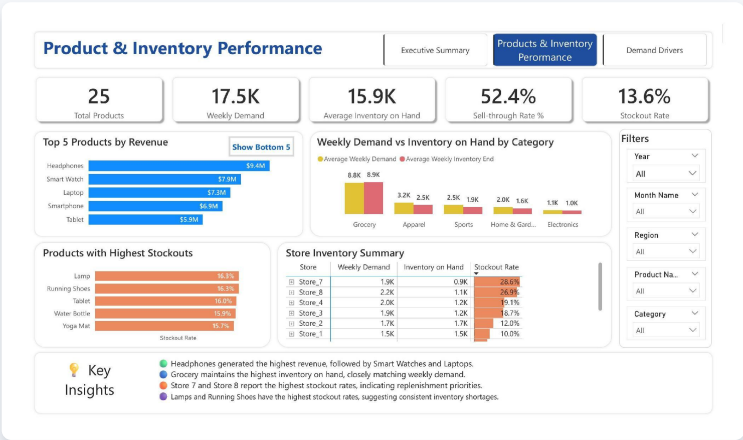
- Sales quantity & unit price
- Inventory start / end levels
- Competitor price
- Promotion & holiday flags
- Stockout flag
- Forecasted demand quantity
- Store, region & category IDs
- Week / date

DATA PREPARATION

Exploratory data analysis (EDA) was performed using Python to understand sales patterns and data quality. The data was then modeled into a Power BI star schema with business KPIs including Revenue, Sell-through Rate, Promotion Lift, and Forecast Accuracy for reporting and analysis.

A Three-Page Interactive Power BI Dashboard

Cross-filtering, drill-down & slicers for Year, Month, Region, Store, Product and Category



Executive Overview

Revenue, units sold, average price, stockout rate and promotion lift at a glance — with trend and regional breakdowns for leadership reporting.



Product & Inventory Performance

Top products by revenue, stockout exposure by store and category, and inventory-vs-demand balance for replenishment planning.



Demand Drivers

Promotion and holiday impact on demand, competitor price positioning, and forecast-vs-actual accuracy for planning teams.

KEY PERFORMANCE INDICATORS

Metrics That Matter to the Business



Revenue

Measures: Total sales value generated

Why it matters: Core measure of business performance and growth

\$82.1M tracked across 10 stores and 5 regions



Units Sold

Measures: Total product volume sold

Why it matters: Shows demand strength independent of pricing

1.84M units sold over 2 years



Stockout Rate

Measures: % of transactions with unmet demand

Why it matters: Flags lost sales and availability gaps

13.6% overall — highlights replenishment priorities



Promotion Lift

Measures: Demand increase during promotions

Why it matters: Quantifies campaign effectiveness

+39.5% average unit lift during promotions



Avg. Weekly Sales

Measures: Typical weekly demand by product/category

Why it matters: Sets a baseline for planning and replenishment

Benchmarks stores and categories consistently



Forecast Accuracy

Measures: How closely forecasts match actual demand

Why it matters: Determines planning and inventory reliability

62.8% accuracy — key improvement opportunity

What the Dashboard Reveals

1

Category Mix Is Skewed

Electronics drives 45.6% of revenue from just 6.1% of units sold, while Grocery delivers 50.1% of volume but only 9.2% of revenue — pricing and merchandising strategy should differ sharply by category.

2

Regional Performance Gap

East is the top-performing region (23.9% of revenue), while Central lags — a signal to investigate what East is doing right and replicate it.

3

Promotions Pay Off, Selectively

Promotions lift average units by 39.5% and holidays by 34.0%, with Grocery (39.9%) and Electronics (48.0% holiday response) responding strongest — a case for targeted, not uniform, campaign spend.

4

Inventory Risk Is Concentrated

1 in 7 transactions faces a stockout (13.6% overall), spiking to 32.1% during promotions — with West/South regions and Store_7/Store_8 carrying the highest exposure.

5

Forecasts Weaken When It Matters Most

Forecast accuracy sits at 62.8%, and error widens sharply during promotions and holidays — precisely when accurate planning is most valuable.

6

Pricing Sweet Spot Identified

Products priced within $\pm 5\%$ of competitor prices achieve the highest average sales volume — supporting close price-matching over aggressive discounting.

From Raw Data to Interactive Insight



1. Data Exploration (Python)

Validated data quality and performed exploratory data analysis (EDA) using Pandas to identify trends and patterns



2. Star Schema Modeling

Designed a star schema in Power BI — one fact table, three dimensions — for fast, reliable, scalable calculations.



3. DAX Measures

Built business-ready DAX measures for KPIs like stockout rate, promotion lift, sell-through rate and forecast accuracy.



4. Interactive Visualizations

Delivered a three-page Power BI dashboard with cross-filtering, drill-down, and slicers for self-service exploration.

CONCLUSION

A Complete, Actionable View of Retail Performance

This project demonstrates a full business-intelligence workflow — from raw retail data through cleaning, exploratory analysis, star-schema modeling and interactive dashboard development, to concrete, actionable recommendations. The result is a single source of truth that helps stakeholders monitor performance, reduce stockout risk, evaluate promotions, and plan inventory with confidence — replacing guesswork with evidence-based decisions.

Skills demonstrated: · Python (Pandas) · Power BI · DAX · Star-Schema Data Modeling · Business Storytelling